CAT Commit Protocol | 1–3

5 prompts · 7 min. Practice the forward-only decision routine; name exact actions.

Date: _____ Show your reasoning in the space provided.

A1 After you submit a standalone Math item, what should you expect to be able to do—and not do?

Think: State both parts precisely.

Read, solve, and check while the item is still on screen.

After submitting and advancing, you cannot reopen the previous

standalone Math item to reread it or change its answer.

Answer: **Advance to the next item; you cannot return to the submitted item.**

A2 A classmate receives a harder next item. Explain why this is not proof that the classmate has already earned a higher final score.

Think: Separate item selection from final performance.

Scoring uses the full pattern of correct and incorrect responses

in relation to item difficulty. The next item is not a score report,

and responses to the remaining items still affect the final estimate.

Answer: **One harder next item does not establish a higher final score.**

A3 You calculated 0.36, but the prompt asks for the numerical value of the percent. What should you type, and what should you omit?

0.36 × 100 = 36, so 0.36 = 36%.

The field requests the numerical value of the percent.

Check: 36 ÷ 100 = 0.36.

Answer: **Type 36; omit the percent sign.**

CAT Commit Protocol | 4–5

Continue the forward-only routine. Target → represent → solve → verify → commit.

Date: _____ Show your reasoning in the space provided.

A4 You have a plausible answer with 25 seconds invested, but your estimate disagrees. Name one fast audit that is worth doing before advancing.

Think: Audit + what the result would tell you.

Multiply the proposed quotient by the original divisor.

The product must equal the original dividend.

For example, 0.14 × 3.9 = 0.546, not 5.46, so 3.9 fails the check.

But 0.14 × 39 = 5.46, which verifies 39.

Correct the scale error and reread the requested form before advancing.

Answer: **Audit place value or scale with one targeted inverse check.**

A5 There are several items left near the end of testing time. Write a two-sentence rule that balances thoughtful attempts with the need to keep moving.

Think: Sentence 1: default action. Sentence 2: observable move-on cue.

I will make a focused best attempt on each remaining item, using a quick model or elimination and checking the requested form. If I have no productive next step and the remaining time requires moving on, I will submit my best-supported response and continue rather than stall or click randomly through everything.

Answer: **Make a focused best attempt; move on using a specific time cue.**

Name: _____

Representation Warm-Up | 1–3

6 questions · 9 min. Write a benchmark, equivalent form, or place-value statement first.

Date: _____ Show your reasoning in the space provided.

B1 Write 0.625 as a fraction in lowest terms.

Answer format: Typed entry

Think: Place value → fraction → reduce

0.625 = 625/1000 = 5/8 (divide top and bottom by 125).

Check: 5/8 = 0.5 + 0.125 = 0.625.

Answer: **5/8**

B2 Write 7/16 as a decimal.

Answer format: Typed entry

Think: Division or equivalent denominator

7/16 = 8/16 - 1/16 = 0.5 - 0.0625 = 0.4375.

Check: 0.4375 × 16 = 7.

Answer: **0.4375**

B3 Write 0.072 as a percent. Enter the number without the percent sign.

Answer format: Typed entry

Think: Explain the scale factor

Percent means per hundred: 0.072 × 100 = 7.2%.

Enter 7.2 without the percent sign.

Check: 7.2 ÷ 100 = 0.072.

Answer: **7.2**

Representation Warm-Up | 4–6

Compare quantities using a shared representation before choosing an answer.

Date: _____ Show your reasoning in the space provided.

B4 Which value is greatest?

- ◯ A. 11/20
- ◯ B. 0.56
- ◯ C. 54%
- ◯ D. They are equal

Think: Convert to one common language

11/20 = 0.55; 54% = 0.54.

0.56 > 0.55 > 0.54, so choice B is greatest.

Answer: **B — 0.56**

B5 Without exact calculation, place 13/24 relative to 1/2 and justify.

- ◯ A. Less than 1/2
- ◯ B. Equal to 1/2
- ◯ C. Greater than 1/2

Think: Use a numerator benchmark

1/2 = 12/24. Since 13 > 12, 13/24 > 12/24 = 1/2.

Thirteen is more than half of 24.

Answer: **C — Greater than 1/2**

B6 Order from least to greatest: 9/20, 0.47, 46%.

Answer format: Order

Think: Show a common representation

9/20 = 0.45 and 46% = 0.46.

0.45 < 0.46 < 0.47. List the values in their original forms.

Answer: **9/20, 46%, 0.47**

Core Accuracy Set | 1–4

12 questions · 22 min. Solve, scan the requested form, record, then commit.

Date: _____ Show your reasoning in the space provided.

C1 Compute 5/12 + 7/18.

Answer format: Lowest-terms fraction

Think: Rename → operate → reduce

Commit check: I checked the denominator and form before advancing.

5/12 + 7/18 = 15/36 + 14/36 = 29/36 (lowest terms).

Answer: **29/36**

C2 Compute 1 5/6 − 7/15.

Answer format: Response

Think: Show renaming and regrouping if needed

Commit check: I tested whether my result is between 1 and 2.

1 + (25/30 - 14/30) = 1 11/30 = 41/30; between 1 and 2.

Answer: **41/30 or 1 11/30**

C3 Compute 8.03 − 2.786.

Answer format: Decimal

Think: Rewrite with aligned decimal places

Commit check: I used a subtraction estimate.

8.030 - 2.786 = 5.244; estimate: 8.0 - 2.8 = 5.2.

Answer: **5.244**

C4 Compute 5.46 ÷ 0.14.

Answer format: One number

Think: Scale both numbers equally

Commit check: I multiplied back to verify.

5.46 ÷ 0.14 = 546 ÷ 14 = 39; check: 0.14 × 39 = 5.46.

Answer: **39**

Core Accuracy Set | 5–8

Treat each item as one forward-only screen. Show an intermediate step.

Date: _____ Show your reasoning in the space provided.

C5 Find 3/5 of 45.

- ◯ A. 15
- ◯ B. 27
- ◯ C. 30
- ◯ D. 75

Think: Explain the meaning of “of”

Commit check: I compared the answer with 45.

Answer: **B — 27; (45 ÷ 5) × 3 = 9 × 3 = 27, which is less than 45.**

C6 A number multiplied by 0.35 equals 8.4. Find the number.

Answer format: One number

Think: Equation → estimate → solve

Commit check: I substituted my value into the equation.

0.35n = 8.4 → 35n = 840 → n = 24; 0.35 × 24 = 8.4.

Answer: **24**

C7 Compute 2.4 + 1 3/8. Give your answer as a decimal.

Answer format: Decimal

Think: Translate one value before combining

Commit check: I returned the requested form.

1 3/8 = 1.375; 2.400 + 1.375 = 3.775.

Answer: **3.775**

C8 What is 17.5% of 80?

Answer format: One number

Think: Choose fraction or decimal method

Commit check: I checked that the answer is less than 80.

17.5% = 7/40; (80 ÷ 40) × 7 = 14, which is less than 80.

Answer: **14**

Name: _____

Core Accuracy Set | 9–12

Check the original quantity and the form requested before committing.

Date: _____ Show your reasoning in the space provided.

C9 A price rises from $48 to $60. What is the percent increase?

Answer format: Number only

Think: Change ÷ original

Commit check: I divided by the original value.

(60 - 48) ÷ 48 = 12/48 = 0.25 = 25%; enter 25.

Answer: **25**

C10 A $72 jacket is discounted 25%. What is the sale price?

Answer format: Dollars

Think: Discount amount or remaining percent

Commit check: I answered sale price, not discount.

Discount = 72 ÷ 4 = 18 dollars; sale price = 72 - 18 = 54 dollars.

Answer: **54**

C11 A quantity is increased by 20% and becomes 54. What was the original quantity?

Answer format: One number

Think: Represent 54 as a percent of the original

Commit check: I did not subtract 20% from 54.

1.2x = 54 → x = 45; check: 45 + 0.20 × 45 = 54.

Answer: **45**

C12 The value of 3/8 of a number is 6 less than 1/2 of the number. Find the number.

Answer format: One number

Think: Write one equation using the same whole

Commit check: I checked both fractional amounts.

(1/2 - 3/8)n = 6 → n/8 = 6 → n = 48; 24 - 18 = 6.

Answer: **48**

SHSAT Transfer Set | 1–3

8 questions · 18 min. Model the situation before choosing an operation.

Date: _____ Show your reasoning in the space provided.

D1 A library shelf is 2/9 empty. After 14 books are removed, it is 1/3 empty. How many books fit on the shelf?

Answer format: Capacity

Think: Before → change → after

Change in empty space: 1/3 - 2/9 = 1/9 of capacity.

(1/9)N = 14 → N = 14 × 9 = 126 books.

Check: 28 empty spaces + 14 = 42; 42/126 = 1/3.

Answer: **126 books**

D2 A container holds 4.8 liters. It dispenses 1 3/5 liters in the morning and 0.95 liter later. How many liters remain?

Answer format: Number only

Think: Use one representation for all amounts

1 3/5 = 1.6 liters. Total dispensed = 1.60 + 0.95 = 2.55.

Remaining = 4.80 - 2.55 = 2.25 liters.

Check: 2.25 + 1.60 + 0.95 = 4.80.

Answer: **2.25**

D3 A rectangle has perimeter 62 units and length 19 units. What is its area?

Answer format: Number only

Think: Find the missing side first

L + W = 62 ÷ 2 = 31; W = 31 - 19 = 12.

Area = 19 × 12 = 228 square units.

Check: 2(19 + 12) = 62. Enter only the number.

Answer: **228**

SHSAT Transfer Set | 4–6

The final answer should contain only the requested quantity.

Date: _____ Show your reasoning in the space provided.

D4 The mean of 8, 13, 16, 21, and x is 15. Find x.

Answer format: One number

Think: Required total − known total

Required total = 15 × 5 = 75.

Known total = 8 + 13 + 16 + 21 = 58; x = 75 - 58 = 17.

Check: (58 + 17) ÷ 5 = 15.

Answer: **17**

D5 Four quiz scores have a mean of 18. A fifth score is added, and the new mean is 20. What is the fifth score?

Answer format: One number

Think: Compare the two totals

Old total = 4 × 18 = 72. New total = 5 × 20 = 100.

Fifth score = 100 - 72 = 28.

Check: (72 + 28) ÷ 5 = 20.

Answer: **28**

D6 On a map, 1.5 inches represents 12 miles. Two towns are 4.25 inches apart on the map. What is the actual distance?

Answer format: Miles

Think: Unit rate or proportion

Unit rate = 12 ÷ 1.5 = 8 miles per inch.

Distance = 4.25 × 8 = 34 miles.

Check: 34 ÷ 8 = 4.25 inches.

Answer: **34 miles**

SHSAT Transfer Set | 7–8

Evaluate every choice independently; explain the misconception and benchmark.

Date: _____ Show your reasoning in the space provided.

D7 Select all expressions equal to 0.24.

- ◯ A. 6/25
- ◯ B. 24%
- ◯ C. 24/10
- ◯ D. 12/50

Think: Verify each selection independently

A: 6/25 = 24/100 = 0.24.

B: 24% = 24/100 = 0.24.

C: 24/10 = 2.4, not 0.24.

D: 12/50 = 24/100 = 0.24.

Select A, B, and D; leave C unselected.

Answer: **A, B, and D**

D8 A student claims that 4/7 + 3/5 = 7/12. Identify the exact misconception and use one benchmark to prove the result is impossible.

Answer format: Correct sum

Think: Misconception + benchmark contradiction

The student added numerators and denominators directly.

Denominators name piece sizes; first use a common denominator.

Benchmark: 4/7 > 1/2 and 3/5 > 1/2, so their sum is greater than 1.

But 7/12 < 1, so the proposed result is impossible.

Correct sum: 4/7 + 3/5 = 20/35 + 21/35 = 41/35.

Answer: **41/35 = 1 6/35**

One-Screen Rehearsal

3 decisions · 5 min. Describe the final check before you advance.

Date: _____ Show your reasoning in the space provided.

E1 Your decimal quotient is greater than the dividend even though you divided by a number less than 1. Is that automatically an error? Explain the size logic.

Think: Decision + mathematical reason

Smaller-than-one groups mean more groups fit into a positive amount.

Example: 6 ÷ 0.5 = 12 half-unit groups.

Check: quotient × divisor must equal the original dividend.

A larger answer is plausible, but this alone does not prove it correct.

Answer: **No; for positive numbers, dividing by 0 < d < 1 increases the value.**

E2 Your rectangle work produced linear units, but the prompt asks for area. What does this reveal about your stopping point?

Think: Name the missing step

You likely stopped at a length or perimeter, not the requested area.

For D3, width = 12 is intermediate; area = 19 × 12 = 228 square units.

If you already multiplied L × W, check whether only the unit label is wrong.

Answer: **Finish by multiplying length × width to find the area.**

E3 Your calculation is correct, but your typed field contains “60 gallons.” State exactly what belongs in the field and why.

Think: Entry + format explanation

Keep units in the reasoning, but follow the response-field directions.

Assuming this is a number-only field, enter 60, not “60 gallons.”

Check that 60 is the requested quantity before advancing.

Answer: **Type 60; omit “gallons” in a number-only field.**

Process Debrief

Evidence, not impressions. Complete before checking solutions.

Date: _____ Show your reasoning in the space provided.

- ☐ I named the requested quantity before calculating.
- ☐ I used a common unit before adding or subtracting fractions.
- ☐ I aligned decimal place values rather than decimal points by habit.
- ☐ I estimated before committing to a decimal quotient.
- ☐ I treated percent change as change divided by the original.
- ☐ I entered only the value and form requested.
- ☐ I made a best attempt instead of rushing through remaining items.
- ☐ I finished every problem without looking back at a prior response.

Timing record

A–B: _____ C: _____ D–E: _____ Total: _____

Confidence audit

List two items you would most want to revisit on a non-adaptive test. Then explain what verification you used instead.

Sample — C11: I was unsure about reversing the percent increase; I verified 45 + 20% of 45 = 54, checking the original quantity rather than subtracting again.

Sample — D1: I was unsure what the 14 books represented; I verified 28 empty spaces before and 42 after, so 14 changes the empty fraction from 2/9 to 1/3.

Error Log

Up to 5 corrections. Identify the first wrong decision, not just the wrong answer.

Date: _____ Show your reasoning in the space provided.

Use one primary type: Concept, Procedure, Reading, Arithmetic, Format, or Wrong Quantity.

Illustrative corrections from the key; not a record of the student's mistakes.

1. Problem: D8 Primary type: Concept

   First wrong decision: Added numerators and denominators: (4 + 3)/(7 + 5).

   Next-time interruption cue: Same-sized pieces? Rename both fractions in thirty-fifths.

2. Problem: C4 Primary type: Procedure

   First wrong decision: Scaled 0.14 to 14 but did not scale 5.46 to 546.

   Next-time interruption cue: Same factor, both numbers: write ×100 beside each.

3. Problem: D3 Primary type: Arithmetic

   First wrong decision: Computed 19 × 2 as 28, giving 218 for 19 × 12.

   Next-time interruption cue: Check a nearby multiple: 20 × 12 - 12 = 228.

4. Problem: C10 Primary type: Wrong Quantity

   First wrong decision: Entered the $18 discount instead of the sale price.

   Next-time interruption cue: Saved or paid? Subtract the discount from the original price.

5. Problem: B3 Primary type: Format

   First wrong decision: Typed the percent sign despite number-only directions.

   Next-time interruption cue: Value and form? Type 7.2 without the percent sign.

Name: _____

Transfer Reflection

Complete sentences. Describe actions that you can repeat on the next problem.

Date: _____ Show your reasoning in the space provided.

1. The representation that prevented my most likely error was...

Model: The representation that prevented my most likely error was dividing the shelf into nine equal blocks in D1, because it showed that the 14 removed books represented the change of one ninth of the capacity, not all the empty space.

2. One answer-format trap I can now recognize is...

Model: One answer-format trap I can now recognize is leaving a decimal unchanged when the prompt asks for the numerical value of a percent; in B3, 0.072 becomes 7.2%, so I enter 7.2 without the percent sign.

3. When an item feels difficult, my next observable action will be...

Model: When an item feels difficult, my next observable action will be to underline the requested quantity and write one equation or diagram connecting it to the givens, instead of trying to infer my score from how hard the item feels.

4. My five-second commit question will be...

Model: My five-second commit question will be: Did I answer the requested quantity, does its size make sense, and is my final response in the required form with every needed selection?

Name: _____